

(Please write your Enrollment Number)

Enrollment No. 01501012021

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: < B.Tech >, Semester: < 6th Sem >
(May, 2024)

Subject Code: BCS 306	Subject: Compiler Design	Maximum Marks :60
Time :3 Hours		
Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.		

Q1	(2.5*8=20)	
(a)	Write about the first phase in Compiler? How a token is recognized, explain with an example?	
(b)	What is the importance of semantic analysis, explain?	
(c)	What is top down parsing? Design Recursive Descent Parser for the following grammar: $S \rightarrow AA$, $A \rightarrow aA$, $A \rightarrow b$	
(d)	Briefly explain about Syntax Directed Translation.	
(e)	What is code optimization and how it is useful in compilation process?	
(f)	Define FIRST and FOLLOW function Rules, explain with an example Grammar.	
(g)	What is the second phase in Compiler? How grammar is useful in second phase?	
(h)	Define Ambiguity? Check whether the grammar $S \rightarrow aAB$, $A \rightarrow bC/cd$, $C \rightarrow cd$, $B \rightarrow c/d$, is Ambiguous or not?	
UNIT-I		
Q2	With a neat diagram discuss about different phases of the compiler? Explain with suitable example.	(10)
Q3	Briefly explain about the following with suitable example: (i) Finite Automata (ii) Derivation tree and parse tree (iii) Ambiguity with an example (iv) Bootstrapping	(10)
UNIT-II		
Q4	Discuss about SLR parser? Construct SLR parse table for following grammar. $E \rightarrow E+T$ $E \rightarrow T$ $T \rightarrow T * F$ $T \rightarrow F$ $F \rightarrow (E)$ $F \rightarrow id$	(10)
Q5	By explaining the steps followed to construct LALR parse table, Design a LALR parse table for the following grammar. $S \rightarrow AA$ $A \rightarrow aA/b$	(10)
UNIT-III		
Q6	Define what Syntax directed definitions is, and explain about following with example. (i). Bottom-up evaluation of S- attributed Definitions (ii). Bottom-up evaluation of inherited attributes.	(10)
Q7	(i). Discuss about different Source Language issues? (ii). Explain about various Storage allocation strategies?	(10)
UNIT-IV		
Q8	Explain about the following with suitable example: (i). Three Address code (ii). Quadruples (iii). Triples (iv). Graphical Representation of Intermediate language.	(10)
Q9	What is code optimization, What are its uses? Explain in detail about Principal sources of optimization with suitable example.	(10)

(Please write your Enrollment Number)

Enrollment No. 01501012021

MID-TERM EXAMINATION
(Course Name: B.Tech. IT/ECE/CSE) (Semester: 6th)
(February-March, 2024) OFF-LINE mode

Subject Code: BIT-304	Subject: Cloud Computing
Time : 1 ½ Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

Q1	Describe the following terms and technologies using suitable illustrations.	(2.5*4)	
	(a) Virtual Provisioning		
	(b) Peer-to-Peer Computing		
	(c) Software Isolation		
	(d) Virtual Storage Containers		
Q2	(Attempt any Two Parts) UNIT-1	(5,5)	
	(a) In a hyperconnected cloud-enabled IGDTUW environment, suggest a suitable Cloud Computing Architecture based on Load Balancing for the IGDTUW Examination portal from the student's perspective.		
	(b) Categorically differentiate between Cloud and Grid Computing frameworks. Describe the advantages and challenges of Grid Computing.		
	(c) Design a hybrid cloud deployment model for the "IGDTUW Taaranga" portal from the frontend perspective. Incorporate the scalability and security compliance factors in the designed architecture.		
Q3	(Attempt any Two Parts) UNIT-2	(5,5)	
	(a) Regarding the Virtualization Reference Model, discuss the salient characteristics of the Virtualization Layer along with a detailed reference model. Highlight the advantages of virtualization and mention the limitations.		
	(b) With reference to Infrastructure-as-a-Service in the Cloud Cube Model, discuss the significance of the Vendor Lock-In problem and its Operational Limitations with suitable illustrations.		
	(c) Regarding the Linthicum Deployment Model, discuss the significance of Process-as-a-Service and Security-as-a-Service with suitable illustrations. Discuss the Cloud Security characteristics associated with Jericho Forum Cloud Cube and Linthicum cloud deployment models.		

(Please write your Enrollment Number)

Enrollment No. 01501012021

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech>, Semester:<6th>
(May, 2024)

Subject Code: BIT-304	Subject: Cloud Computing
Time :3 Hours	Maximum Marks :60
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

Q1	Explain the following terms with suitable illustrations.	(2.5*8=20)	
	(a) Cloud Software Testing		
	(b) Client Information Privacy		
	(c) VM Security		
	(d) Network Intrusion		
	(e) Dynamic Computing Infrastructure		
	(f) Failure Detection and Recovery		
	(g) Information Life Cycle in Cloud Computing		
	(h) Elastic Computing		
UNIT-I			
Q2	Critically analyze the benefits and limitations of distributed and decentralized computing in Web 3.0.	(10)	
Q3	Discuss the Data Lock-In and Data-Segregation limitations in Cloud Computing. State the effective measures to overcome the aforementioned problems in cloud environment.	(10)	
UNIT-II			
Q4	In the realm of Virtualization at Server in Cloud Computing, differentiate between Logical Partitioning and Physical Partitioning with suitable illustrations.	(10)	
Q5	Quantitatively differentiate between Virtualization at Server, at Network and at Desktop and Application in Cloud Computing paradigm. Explain with suitable illustrations.	(10)	
UNIT-III			
Q6	Critically analyze the Cloud Information Security objectives from clients' perspective. Simultaneously highlight the Cloud Security Design Principles from organizations' perspective.	(10)	
Q7	Categorically differentiate between Internet-based Software Testing and Cloud Testing with suitable real-world examples.	(10)	
UNIT-IV			
Q8	Quantitatively assess the crucial factors adopted by Govt. of India in adopting a specific Cloud Computing Architecture for their e-Healthcare Management System.	(10)	
Q9	Highlight the significance of Service-Level-Agreements between the cloud service provider and its client. Enlist the SLAs' salient characteristics along with its privacy concerns.	(10)	

(Please write your Enrollment Number)

Enrollment No. 01501012021

MID-TERM EXAMINATION
(Course Name: B.Tech IT/CSE/MAE/DMAM/CSE-AI) (Semester 6)
(March, 2024) OFF LINE mode

Subject Code: HMC 306	Subject: Financial Management
Time : 1 ½ Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

Q1		(2.5*4)	
	(a) An investor expects a perpetual sum of ₹500 annually from his investment. What is the present value of this perpetuity if the rate of interest is 10%?		
	(b) Mr. B invested ₹1,00,000 at the beginning of one year at the rate of 10% compounded annually. What amount of money he will receive at the end of second year of his investment?		
	Table values: When $r=10\%$, $n=2$ PVIF- 0.826 , PVIFA- 1.736 FVIF- 1.210 , FVIFA- 2.100		
	(c) Suppose a firm has very high business risk, should the firm go for more debt financing. Support your opinion with reason.		
	(d) Highlight few features of preference shares.		
Q2	(Attempt any Two Parts) UNIT-1	(5,5)	
	(a) A company wants to replace an existing machinery after five years. The machinery is likely to cost ₹5,80,000 at the end of five years. However 80,000 is the scrap value of this machinery at the end of five years. Find out annual amount of money to be deposited at 10% rate of interest for next five years to get the desired sum of money for replacement of this machinery.		
	Table values: When $r=10\%$, $n=5$ PVIF- 0.621 , PVIFA- 3.791 FVIF- 1.611 , FVIFA- 6.105		
	(b) Wealth maximisation is superior criteria than profit maximisation. Do you agree, explain with example.		
	(c) Future value of a cash flow is different from its present value today. Justify with the help of a suitable example.		
Q3	(Attempt any Two Parts) UNIT-2	(5,5)	
	(a) Suppose you are a finance manager of a new start up, discuss the factors determining the optimum capital structure of this firm.		
	(b) Discuss significance of cost of capital for any firm.		
	(c) What do you mean by explicit cost and implicit cost? How they both differ from each other?		

(Please write your Enrollment Number)

Enrollment No. 01501012021

End-Term Examination
(CBCS)(SUBJECTIVE TYPE) (Offline)
Course Name:<B.tech>, Semester:< 6 >
(May, 2024)

Subject Code: HMC 306

Subject: Financial Management

Time :3 Hours

Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1

(2.5*8=20)

- (a) What do you mean by financial management?
- (b) Discuss ordinary shares.
- (c) What are the limitations of pay back period?
- (d) Differentiate between temporary working capital and permanent working capital.
- (e) ABC Ltd. expects its cost of goods sold for 2022-2023 to be rs 600 lakh. The expected operating cycle is 90 days. It wants to keep a minimum cash balance of 10,000. What is the expected working capital requirement. Assume a year consists of 360 days.
- (f) Why is it possible for IRR and NPV method to result in different ranking of investment proposal?
- (g) What are mutually exclusive investment projects?
- (h) Write a short note on capital rationing?

UNIT-I

Q2

- A) "Traditional financial management was concerned with the raising of funds as well as their optimum utilisation". Do you agree? Explain.
- B) Mr X is supposed to get ₹10,000 after three years. The existing rate of interest is 10%, find the present value of this sum
- Table values: $r=10\%$, $n=3$
- PVIFA - 2.487 PVIF- 0.751 (7+3)

(10)

Q3

What are the major types of financial management decisions that business firm make? Describe briefly each one of them.

(10)

UNIT-II

Q4

Explain the different types of cost of capital with examples in brief.

(10)

Q5

Write a detailed note on capital structure.

(10)

UNIT-III

Q6	A Choice is to be made between two proposals which require an equal investment of ₹50,000 each and are expected to generate net cash (CFAT) flows as under: <table><tr><th>Year</th><th>Project A</th><th>Project B</th></tr><tr><td>1</td><td>125,000</td><td>10,000</td></tr><tr><td>2</td><td></td><td>15,000</td></tr><tr><td>3</td><td>310,000</td><td>12,000</td></tr><tr><td>4</td><td></td><td>18,000</td></tr><tr><td>5</td><td>4nil 25,000</td><td></td></tr><tr><td>6</td><td>512,000</td><td>8,000</td></tr><tr><td>7</td><td>6 6,000</td><td>4,000</td></tr></table> The cost of capital of the company is 10%. Which project should be chosen under: a) payback period b) NPV method PVIF @ 10%- 0.909, 0.826, 0.751, 0.683, 0.621, 0.564 PVIFA @ 10%-0.909, 1.736, 2.487, 3.170, 3.791, 4.355	Year	Project A	Project B	1	125,000	10,000	2		15,000	3	310,000	12,000	4		18,000	5	4nil 25,000		6	512,000	8,000	7	6 6,000	4,000	(10)
Year	Project A	Project B																								
1	125,000	10,000																								
2		15,000																								
3	310,000	12,000																								
4		18,000																								
5	4nil 25,000																									
6	512,000	8,000																								
7	6 6,000	4,000																								
Q7	A) A company is considering an investment proposal to install a new machinery. The machinery will cost ₹50,000 with the Life of 5 years and no salvage value. The tax rate of the company is 55% and it uses straight line method of depreciation. The estimated cash flows before tax and depreciation are as follows: <table><tr><th>YEAR</th><th>PBDT</th></tr><tr><td>1</td><td>10,000</td></tr><tr><td>2</td><td>11,000</td></tr><tr><td>3</td><td>14,000</td></tr><tr><td>4</td><td>15,000</td></tr><tr><td>5</td><td>25,000</td></tr></table> Calculate: a) Net present value at 10% discount rate b) Profitability index at 10% discount rate PVIF @ 10%- 0.909, 0.826, 0.751, 0.683, 0.621 PVIFA @ 10%-0.909, 1.736, 2.487, 3.170, 3.791	YEAR	PBDT	1	10,000	2	11,000	3	14,000	4	15,000	5	25,000	(10)												
YEAR	PBDT																									
1	10,000																									
2	11,000																									
3	14,000																									
4	15,000																									
5	25,000																									
UNIT-IV																										
Q8	ABC Ltd purchases 9000 units of spare parts annually, ordering one month usage at a time. Each spare part cost ₹20. The ordering cost per order is ₹15 and the carrying charges are 15% of unit cost. Do you think , It is a correct policy? If not, suggest a more economical purchase policy for the company. Also find out the total cost of inventory ordering and carrying. (Use mathematical method)	(10)																								
Q9	A) The cost per unit of inventory is ₹16. Carrying cost is 15% of average inventory for annum. The ordering cost is 960. The number of unit required is 2000 per year. You Are required to calculate: a) EOQ (using mathematical method) b) Total annual inventory cost at EOQ B) Why do firms hold inventories? (6+4)	(10)																								

MID-TERM EXAMINATION
(Course Name : B. Tech CSE) (Semester : 6)
(March, 2024) OFF LINE mode

Subject Code: BCS- 304	Subject: Microprocessor and Interfacing
Time : 1 ½ Hours	Maximum Marks : 30
Note: Attempt all Questions.	

Q1	Attempt all parts.	(2.5*4)	
	(a) Differentiate Microprocessor and Microcontroller	UNIT-1	
	(b) In 8085 explain the function of the following pins: HOLD, RESET OUT, READY, ALE, TRAP	UNIT-1	
	(c) Explain all four rotate instructions in 8086 microprocessor.	UNIT-2	
	(d) Calculate effective address for: MOV[BX+DI+2], CL if before execution CL= FFH, DS= 0500H, BX=0050H, DI=0007H.	UNIT-2	
Q2	(Attempt any Two Parts) UNIT-1	(5,5)	
	(a) Explain the function of the following pins in 8086: i) DT/R, ii) RESET, iii) LOCK, iv) DEN, v) BHE		
	(b) Why is the 8086 memory organized into two banks of even and odd addresses? How is bank selection achieved using BHE and A ₀		
	(c) What is pipeline flushing? When does it occur in 8086 system? Explain with the help of an example how pipeline flushing affects the system performance.		
Q3	(Attempt any Two Parts) UNIT-2	(5,5)	
	(a) Draw bus cycle timing diagram for the memory read and memory write in minimum mode. List the activities in each clock cycle.		
	(a) Consider following given values: i) DS=0102H, DI=2212H, AX=0101H Find effective address and the content of AX, DI and DS after the execution of following instruction: LDS AX, [DI] ii) SS=0100H, DS=2000H, ES=1002H, BX=1001H, BP=2100H, SI=0002H Find effective address and content of BX, BP, SI, ES, SS after the execution of following instruction: LES BX, [BP]+[SI]		
	(b) Explain following directives in 8086: ASSUME, PROC and ENDP, PUBLIC, SEGMENT and ENDS, ORG		

(Please write your Enrollment Number)

Enrollment No. 171501012021

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name:<BTech CSE>, Semester:<6th>
(May, 2024)

Subject Code: BCS 304	Subject: Microprocessor and Interfacing
Time :3 Hours	Maximum Marks :60
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

Q1		(2.5*8=20)	
	(a) How is it helpful for a processor if it has a large number of internal registers		
	(b) Explain the requirement of Program counter Stack pointer and Instruction Register in 8085		
	(c) Distinguish between Far and Near Jump		
	(d) What do CMP AX,BX do		
	(e) Differentiate between Interrupt service routine and Interrupt Vector		
	(f) Differentiate between Symmetrical and Asymmetrical processing		
	(g) Explain the merits and demerits of Synchronous and Asynchronous data communication		
	(h) Explain Cycle stealing DMA operation		
UNIT-I			
Q2	a) Explain the function of the following pins in 8085: READY, SOD, ALE, RESET OUT, HLDA b) Discuss the function of the address bus and the direction of the information flow on the address bus. How many address lines are necessary to address 2MB(2048K) of memory?	(5,5)	
Q3	a) Describe the function of the 8086 queue. How does the queue speed up the processing? b) Find the result for each line of this program segment(write all values of the answer in binary) STC MOV AX, 5485H RCR AL,1 MOV CL,03 RCL AX,CL MOV CL,05 ROR AX,CL ROL AX,1	(5,5)	
UNIT-II			
Q4	a) Write a program to: i) Store 5B16H at the top of the stack and store the same data to BX register from memory. ii) Rotate the content of the AX register through carry left by 03 bits. b) How assembler directives are different from instructions? Explain the following: ASSUME, LOOP, ENDP, SAHF	(5,5)	
Q5	a) Draw flag register of 8086. Find the result and state of the flags CF, ZF, and OF due to the following instructions. Given AX=008CH, BX=345EH, CX=67EBH i) AND BL,CL ii) OR AH,BH iii) XOR AL,CH	(5,5)	

	iv) TEST AH,BL b) What will be the content of AX and DX after the following operations: i) MOV DX,0000H MOV AX, 222H MOV CX,100H MUL CX ii) MOV AX,1234H MOV DX,0000H MOV BX, 0010H DIV BX iii) MOV AX, 4000H MOV DX, 500H MOV BX, 0010H DIV BX		
UNIT-III			
Q6	a) Explain 8086 interrupts and in context to 8086 interrupts compare the following: i) Edge triggered and Level triggered ii) vectored and non-vectored iii) hardware and software iv) maskable and nonmaskable b) What is a Multiprocessing environment? In the context of multiprocessing systems explain the following terms: Fault-tolerant systems, Reliability, Throughput, Graceful degradation	(5,5)	
Q7	a) Explain and differentiate Loosely coupled and Tightly coupled systems. b) What happens when a number of interrupts occur at the same time? How does the microprocessor handle it? Draw the interrupt vector table of 8086, what is the size of the vectored address in 8086, and explain how the interrupt vector address is calculated in the vector table	(5,5)	
UNIT-IV			
Q8	a) Explain USART in brief with the help of a block diagram. Write mode control word for asynchronous transmission with 8-bit data format, one stop bit, odd parity, and 10000 baud rate approximately. The TxC is 150 KHz. b) Give the sequence of operation of the DMA mode of data transfer; explain in detail. In what situation can the DMA mode of data transfer be used	(5,5)	
Q9	a) Compare 8086, 80186, and 80286 microprocessors. What is paging in which microprocessor it was introduced b) Briefly explain how 8259 channels and interrupt requests from a peripheral to 8086. Discuss the roll of the following in handling multiple interrupts IPR, IMR, ISR	(5,5)	

MID-TERM EXAMINATION
(Course Name: B. Tech) (Semester: 6th Semester)
(March, 2024) OFF LINE mode

Subject Code: BCS 310	Subject: Human Computer Interaction
Time : 1 ½ Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

Q1		(2.5*4)	
	(a) Explain the fundamental components essential for Human-computer interaction.	UNIT-1	CO ₁
	(b) Discuss the differences between UI and UX, illustrating each concept with specific examples to demonstrate understanding.	UNIT-1	
	(c) What is the significance of human interaction speeds in computer interface design?	UNIT-2	CO ₂
	(d) Analyze the complete interaction process between human and machine, utilizing a real-life example to illustrate the sequence of action and feedback involved in the user-machine interaction along with a diagram.	UNIT-2	
Q2	(Attempt any Two Parts) UNIT-1 (CO₁)		(5,5)
	(a) Discuss in detail the various types of user interfaces, providing a comprehensive explanation of each.		
	(b) Explain the general principles guiding the design of user interfaces, illustrating each principle with relevant examples.		
	(c) Compare and contrast direct and indirect manipulation, providing key distinguishing points for each method.		
Q3	(Attempt any Two Parts) UNIT-2 (CO₂)		(5,5)
	(a) Identify and discuss the constraints and limitations specific to mobile and handheld platforms contrasted with other computing platforms.		
	(b) Explain the significance of human characteristics contributing to the design of computer interfaces?		
	(c) Discuss the principles and guidelines that should be adhered to when designing conceptual models for identifying fundamental business functions?		

(Please write your Enrollment Number)

Enrollment No. 01501012021

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: B.Tech, Semester: 6th Semester
(May, 2024)

Subject Code: BCS 310	Subject: Human Computer Interaction
Time : 3 Hours	Maximum Marks : 60
Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

		(2.5*8=20)	
Q1			
(a)	Assess the reasons behind the popularity of web user interfaces in comparison to traditional interfaces, providing two critical factors.		
(b)	Explain the concept of "direct manipulation" in graphical user interfaces and its impact on user experience.		
(c)	Define the term "human interaction with computers" and provide proper examples.		
(d)	Explain the concept of human interaction speeds and its significance in computing.		
(e)	What are some examples of distractions that can affect screen users?		
(f)	Evaluate the effectiveness of grouping techniques using borders in enhancing screen organization.		
(g)	Assess the significance of voice-activated commands in modern user interfaces. How do they enhance user experience?		
(h)	What are the challenges associated with natural language specifications? How do ambiguities in natural language specifications affect communication and implementation?		
UNIT-I			
Q2	Combine the characteristics and principles of user interface design to propose guidelines for creating effective web user interfaces. Provide a rationale for each guideline and explain how it contributes to a positive user experience.	(10)	
Q3	Define the term "user interface" and provide examples of common interfaces in everyday life.	(10)	
UNIT-II			
Q4	Compare and contrast the limitations and constraints of mobile and hand-held platforms with those of desktop computers. Highlight specific issues related to network connectivity, power consumption, memory management, and UI responsiveness, providing examples for each.	(10)	
Q5	Explain why considering human characteristics is crucial in designing user interfaces for various computing platforms. Provide real-world examples to support your explanation.	(10)	
UNIT-III			
Q6	What are the key elements and qualities of visually pleasing composition? Create simple info-graphics demonstrating the principles of visually pleasing composition, using appropriate images and design elements.	(10)	
Q7	Differentiate between device-based and screen-based controls, providing examples of each.	(10)	
UNIT-IV			
Q8	What are the primary features of interface-building tools? Assess the effectiveness of interface-building tools in facilitating efficient collaboration among development teams.	(10)	
Q9	Identify the layers involved in Interface Development and briefly explain the function of each layer with proper Comparison across the different layers of Interface Development in terms of their roles and functionalities.	(10)	

(Please write your Enrollment Number)

Enrollment No. 01501012021

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech>, Semester:<6>

(May, 2024)

Subject Code: BCS 302	Subject: WIRELESS NETWORKS
-----------------------	----------------------------

Time :3 Hours

Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8 =20)	
	a) Explain the purpose and goals of the IEEE 802.11 standard.		
	b) Describe the concept of localization in WSNs. Explain with the help of an example.		
	c) Define unicast routing and multicast routing and explain its purpose.		
	d) Discuss the advantages of using mesh networks over traditional network architectures.		
	e) Discuss the impact of environmental factors (such as interference, noise, and fading) on WSN performance.		
	f) What advantages do Wireless Mesh Networks offer in terms of coverage, scalability, and fault tolerance?		
	g) How do hybrid protocols combine features of distance-vector and link-state routing		
	h) Differentiate between indoor and outdoor adhoc mobility models		

UNIT-I			
Q2	In the context of MAC protocols, define the term contention based protocols and how can you classify contention based protocols? Explain one protocol of each classification in detail.	(10)	
Q3	What are the defining characteristics, key features, and notable applications of ad-hoc networks, and how do these self-configuring networks differ from traditional infrastructure-based networks in terms of their structure, functionality, and deployment scenarios?	(10)	
UNIT-II			
Q4	What is the role and functionality of hierarchical routing protocols in wireless networks? Discuss the specific architectural components and their interactions within hierarchical routing schemes in wireless networks, highlighting their contributions to network scalability, efficiency, and resilience. Explain with the help of an example.	(10)	
Q5	How do ad-hoc transport protocols differ from traditional transport protocols in wired networks, and what unique challenges do they address in dynamic and self-configuring wireless environments? Can you compare and contrast the performance of different ad-hoc transport protocols, such as UDP-based protocols and TCP variants, in terms of their suitability for different types of ad-hoc network applications?	(10)	
UNIT-III			
Q6	Compare and contrast the characteristics and advantages of infrastructure-based and client-based Wireless Mesh Networks (WMNs), highlighting their suitability for different deployment scenarios and network architectures.	(10)	
Q7	What is opportunistic routing, and how does it differ from traditional routing approaches in wireless networks? Discuss the fundamental principles behind opportunistic routing and how it leverages the broadcast nature of wireless communication to improve network performance.	(10)	
UNIT-IV			
Q8	How does the choice of routing protocol affect network performance in WSNs? Compare proactive (table-driven) and reactive (on-demand) routing protocols.	(10)	
Q9	What is the purpose of clustering in WSNs, and how does it impact energy efficiency?	(10)	